

AIM2 Preparation

Week 2 - Day 5: Data Visualization with Matplotlib

5-Minute Review Notes | Matplotlib Fundamentals

Day 5 Goal

Turn Pandas analysis results into clear visualizations and choose the right chart for the question being asked.

Core Workflow

Question -> Pandas analysis -> Choose chart -> Plot -> Label -> Interpret

Matplotlib Basics

Standard import:

```
import matplotlib.pyplot as plt
```

- `plt.figure(figsize=(10, 5))` - creates a figure and controls its size.
- `plt.title(...)` - adds a meaningful chart title.
- `plt.xlabel(...)` / `plt.ylabel(...)` - label the axes.
- `plt.show()` - displays the completed figure.
- `plt.xticks(rotation=45)` - rotates category labels when they overlap.
- `plt.grid()` - adds reference lines when they improve readability.
- `plt.legend()` - identifies multiple plotted series.

The Four Main Charts

Chart	Best Use	Day 5 Example
Line	Trends or ordered changes	Math, Python and AI across students
Bar	Compare categories	Student averages / average Math by city
Histogram	Distribution of one numerical variable	Math score distribution
Scatter	Relationship between two numerical variables	Python vs AI scores

1. Bar Chart - Compare Categories

A bar chart is appropriate when the x-axis contains separate categories, such as students or cities.

```
plt.bar(df["Name"], df["Average"])
plt.xlabel("Students")
plt.ylabel("Average Score")
plt.title("Student Average Scores")
plt.show()
```

Key idea: student names are categories, so bars make comparison clearer than connecting them with a line.

2. Histogram - Understand a Distribution

A histogram groups numerical values into intervals called bins and counts how many observations fall inside each interval.

```
plt.hist(df["Math"], bins=5)
```

What does bins=5 mean? Matplotlib divides the numerical range of Math scores into 5 intervals and counts the scores in each interval.

Example: if scores range from 65 to 95, the total range is 30. With 5 bins, each interval is roughly 6 points wide (exact boundaries are chosen by Matplotlib).

Bins	Interval Width	Effect
Few bins	Wider	More values grouped together; less detail
More bins	Narrower	Fewer values per interval; more detail

Important: Changing the number of bins does NOT change the original scores. It only changes how the histogram groups and displays them.

Also remember: x-axis tick labels are not necessarily the exact bin boundaries. Do not infer precise bin ranges only from the visible ticks.

3. Scatter Plot - Study Relationships

Scatter plots are useful when both variables are numerical and we want to see whether they move together.

```
plt.scatter(df["Python"], df["AI"])
plt.xlabel("Python Score")
plt.ylabel("AI Score")
plt.title("Python vs AI Scores")
plt.grid()
plt.show()
```

In today's dataset, the points generally move from bottom-left to top-right, visually supporting the strong positive correlation found during Day 4.

Correlation reminder: a strong visual or numerical correlation does not prove causation.

4. Multiple Line Plot - Compare Series

```
plt.plot(df["Name"], df["Math"], marker="o", label="Math")
plt.plot(df["Name"], df["Python"], marker="o", label="Python")
plt.plot(df["Name"], df["AI"], marker="o", label="AI")
plt.legend()
plt.show()
```

marker='o' makes each observation visible. label=... names each series, and plt.legend() shows which line represents which subject.

5. Pandas Analysis -> Matplotlib

```
city_average = df.groupby("City")["Math"].mean()
```

```
plt.bar(city_average.index, city_average.values)
plt.xlabel("City")
plt.ylabel("Average Math Score")
plt.title("Average Math Score by City")
plt.show()
```

Here:

- city_average.index contains the city names: London, Ottawa, Toronto.
- city_average.values contains the corresponding average Math scores.
- Pandas prepares/aggregates the data; Matplotlib visualizes the result.

Chart Selection Cheat Sheet

Question	Choose
Compare student averages	Bar chart
Show Math score distribution	Histogram
Compare Python and AI relationship	Scatter plot
Show ordered/trend-like changes	Line chart
Compare average Math across cities	Bar chart

Day 5 Mini-Project

Student Performance Visualization Dashboard

1. Student Average Scores - bar chart
2. Math Score Distribution - histogram

3. Python vs AI - scatter plot
4. Subject Performance - Math/Python/AI line comparison
5. Average Math by City - grouped Pandas result shown as a bar chart

Key Takeaways

- Choose a chart based on the analytical question, not just because a plotting function is available.
- Bar = categories; histogram = numerical distribution; scatter = relationship; line = trends/ordered changes.
- Histogram bins are numerical intervals. More bins give narrower intervals and more detail; fewer bins give wider intervals and more grouping.
- Labels, titles and legends make a visualization understandable.
- EDA and visualization work together: analyze with Pandas, visualize with Matplotlib, then interpret the result.

Progress

Week 2 Day 5 - Matplotlib Visualization: COMPLETE

Next: Week 2 Day 6 - continue the visualization sequence.